

Joao Vitor Pires Henares

Technical Design · 3C & Character Controllers · Context-Driven Systems Architecture · Gameplay Systems

Portfolio: joaohenares.com (more projects & prototypes) | E-mail: jvphenares@gmail.com

TECHNICAL SKILLS & TOOLS

Software: Unreal Engine (C++, Blueprints), Unity, Perforce

Programming: C++, C#, Python, Go, TypeScript

Technical Design: 3C & character controllers, systems architecture (event/data/context-driven), design patterns

Languages: Portuguese (native), English (fluent), Spanish (advanced), German (intermediate)

PROJECTS & EXPERIENCE

Mythbound | Technical Designer — Sept 2026 – Present

Competitive PvP Arena-Brawler · Unreal Engine 5 · 18-Person FIEA Capstone

- Joined the project one month ago and rapidly onboarded into a large, unfamiliar **UE5 codebase** to ship production-critical systems ahead of the game's **Steam launch**.
- Implemented **Steam SDK integration**, including username retrieval and a friend-invite system for social play.
- Built a **telemetry pipeline with PlayFab** tracking live gameplay metrics — objective pass rate, ability use/hit rate, and player density heat maps — to inform meta balance.

Ravenous Sky | Technical Designer — Nov 2025 – Jul 2026

18-Person Adventure Game · Unreal Engine 5 · Wanderlust Studios, FIEA Capstone

- Implemented **3C character controller systems** in C++: gradual wind force, player-driven counter-force bracing, and graduated speed response — owning moment-to-moment player feel.
- Designed a performance-optimized **debris system**: object pooling, data-driven configuration, and a parry-shield mechanic.
- Built a C++ **predictive-collision system** using real-time trajectory prediction to detect incoming hazards and trigger time-dilation feedback.

338th Training Squadron, U.S. Air Force | Team Lead — Jan 2026 – May 2026

TaskForce Productivity RPG App · Unity · 7-Person Team

- Led a **7-person multidisciplinary squad** from concept to delivery, owning feature planning, sprint coordination, and stakeholder presentations.
- Designed and implemented core systems — **progression, faction leaderboards, character customization** — and built internal tooling for pipeline consistency.

RACHA | Solo Design + Programming — 6-Week Gauntlet Production

Crime Roguelike Deckbuilder · Unity · Solo

- Shipped a Balatro-inspired roguelike with 25+ agents, 10 card types, and 5 boss modifiers, **surviving every bi-weekly elimination cut** against multi-person teams.
- Built a **strategy-pattern architecture** where new content is addable via new files with zero changes to existing systems, plus a **typed, scoped context system** decoupling logic and a sequential event queue for complex chain reactions.

2DBonk | Solo Design + Programming — 3-Week Production

Award-Winning Survivor Roguelike · Unity · Solo

- Shipped a polished prototype in 3 weeks; awarded **"Most Addictive Game"** at the FIEA Industry Review.
- Designed 10 enemy types, 3 characters, and a full progression system, with custom state machines and a **pooled spawning system** supporting 1,000+ concurrent enemies at ~90fps.

BTG Pactual | Software Engineering Intern — Summer 2023, Summer 2024

São Paulo, Brazil

- Built an **AI-powered document-processing pipeline** that reduced manual extraction work by 90%.
- Developed full-stack features using **TypeScript, React, and Go**, following clean architecture practices.

EDUCATION

Master's in Interactive Entertainment | FIEA, University of Central Florida — Graduating 2026

Specialization: Technical Design · GPA 4.0

B.S. in Computer Science | University of Notre Dame — Graduated 2024

GPA 3.82 · 4× Dean's List